

Chiller Plant Optimisation & HVAC Retrofit - Technical Report

Keppel Bay Tower · Office · Singapore

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1. Executive summary

Cooling is the largest energy end-use at Keppel Bay Tower, accounting for 54% of building electricity in the 2023 baseline year. The water-cooled chilled-water plant (three 650 RT centrifugal chillers, one 300 RT duty chiller and four cooling towers) was sequenced manually by operators using fixed time schedules, and ran at an annual average plant efficiency of 0.86 kW/RT.

Between March and October 2024 the plant was retrofitted with an AI-driven optimisation layer, variable speed drives and smart HVAC monitoring. Over the first twelve months of operation the measured plant efficiency improved to 0.59 kW/RT, cooling energy fell 31% and whole-building energy use fell 18%, with no increase in tenant thermal-comfort complaints.

2. Baseline conditions

- Chiller staging was decided by operators from fixed schedules; chillers often ran at 30-40% part load, where their efficiency is poorest.
- Chilled-water supply temperature was fixed at 6.5°C all year, regardless of cooling load.
- Condenser water pumps and cooling tower fans ran at constant speed.
- Plant efficiency was only spot-checked monthly; there was no permanent measurement and verification (M&V) instrumentation.

Baseline KPI (2023)	Value
Annual building electricity	9.8 GWh
Share of electricity used for cooling	54%
Chiller plant efficiency (annual average)	0.86 kW/RT
Peak cooling load	1,450 RT

3. Measures implemented

The programme combined five measures, delivered in phases so that each could be verified before the next was enabled.

- AI-driven chiller plant optimisation: a machine-learning model forecasts the cooling load 24 hours ahead from weather forecasts and turnstile occupancy data. Every 15 minutes the optimiser selects the chiller combination and the chilled-water and condenser-water setpoints that minimise total plant kW while meeting the load.
- Chilled-water supply temperature reset: the setpoint now floats between 6.5°C and 8.5°C using trim-and-respond logic driven by AHU valve positions, raising chiller efficiency at part load.
- Variable speed drives on condenser water pumps and cooling tower fans, with the condenser-water temperature optimised against outdoor wet-bulb conditions.
- AHU fan static-pressure reset and CO2-based demand-controlled ventilation.
- Smart HVAC monitoring: fault detection and diagnostics rules run on building management system (BMS) trend data and flag stuck valves, simultaneous heating and cooling, sensor drift and short-cycling chillers.

4. Instrumentation and M&V

Permanent high-accuracy instrumentation was installed before any control change: electromagnetic flow meters, temperature sensors accurate to $\pm 0.05^\circ\text{C}$ and power meters on every chiller, pump and cooling tower fan. The same instrumentation calculates plant efficiency in real time and supplies the data for measurement and verification of savings.

5. Commissioning approach

The optimiser first ran for four weeks in shadow mode: it recommended staging and setpoints, and operators approved or rejected each recommendation. Closed-loop control was enabled only after more than 95% of recommendations were being accepted. Operators can override the optimiser from the BMS at any time.

6. Results (first 12 months)

KPI	Baseline	After retrofit
Chiller plant efficiency (kW/RT)	0.86	0.59
Cooling energy (GWh/yr)	5.3	3.7
Building electricity (GWh/yr)	9.8	8.0
Hot complaints per month	14	13
Carbon avoided (tCO ₂ e/yr)	-	725

7. Lessons learned

- Sensor accuracy is the foundation. Two of fourteen chilled-water temperature sensors had drifted by more than 0.3°C and caused poor staging decisions in the first month; quarterly sensor calibration is now mandatory.
- Shadow mode built operator trust. Operators who had run the plant for years accepted closed-loop control because they had seen, and could veto, every recommendation.
- Coordinate temperature reset with humidity. During the wet north-east monsoon the lobby AHUs needed a lower chilled-water temperature for dehumidification, so the reset logic now includes a space dew-point limit.
- Phase the changes. Enabling one measure at a time made savings attributable and rollback simple.

8. Applicability

The approach applies to any water-cooled chilled-water plant. Facilities with steady, round-the-clock cooling loads can expect a larger absolute saving from supply-temperature reset and optimised staging, provided equipment and contractual temperature limits are respected.